

AI QUICK REVISION NOTES

Your 10-Minute Guide to AI Mastery



AI
FUNDAMENTALS



PROMPT
ENGINEERING



MODELS &
TRANSFORMERS



LLMS &
GENERATIVE AI



ETHICS &
FUTURE

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AI Quick Revision Notes

Interview & Exam Cheat Sheet

Goal: Understand AI concepts in 15 minutes.

AI Roadmap

Artificial Intelligence (AI)

|



Machine Learning (ML)

|



Deep Learning (DL)

|



Generative AI

|



Large Language Models (LLMs)

|



ChatGPT • Gemini • Claude

What is AI?

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, decision-making, and problemsolving.

Easy Remember

AI = Making computers think and solve problems like humans.

AI vs ML vs DL

AI

Broad field

Rule + Learning

Smart systems

ML

Learns from data

Pattern finding

Prediction

DL

Uses neural networks

Complex pattern learning

Images, Speech, LLMs

Generative AI

Generative AI creates **new content** instead of only analyzing data.

It can generate:

- Text
- Images
- Code
- Music
- Videos

Examples

- ChatGPT
- Gemini
- Claude
- Midjourney

Types of AI

Narrow AI

Designed for one task.

Example:

- ChatGPT
- Siri
- Google Translate

General AI (AGI)

Can perform almost every intellectual task like a human.

Doesn't exist yet.

Super AI

AI smarter than humans.

Currently theoretical.

Interview Question

Q: Which type of AI is ChatGPT?

Narrow AI

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Prompt Engineering

What is Prompt Engineering?

Prompt Engineering is the skill of giving clear instructions to AI for better results. Better

Prompt = Better Output

CRAFT Framework

C → Context

Tell AI why you need the answer.

Example

Explain AI.

Explain AI for a beginner preparing for interviews.

R → Role

Assign AI an expert role.

Example

"You are an AI Engineer."

A → Action

Tell AI exactly what to do. Examples

- Explain
 - Summarize
 - Compare
 - Design
 - Write
 - Analyze
-

F → Format

Choose output style.

Examples

- Bullet Points
 - Table
 - Notes
 - PDF
 - Flowchart
-

T → Tweaks

Mention extra requirements.

Examples

- Keep it simple
 - One-page summary
 - Professional tone
 - Add examples
-

Prompt Formula

Context



Role



Action



Format



Tweaks

Good Prompt

You are an AI professor.

Explain Transformers.

Use simple English.

Give examples.

Keep under 200 words.

Bad Prompt

Explain AI.

Too short.

No context.

Remember

AI cannot read your mind.

The more specific your prompt, the better the response.

AI Models (Most Important)

What is an AI Model?

A model is a trained program that learns patterns from data and makes predictions or generates content.

Example

GPT is an AI model.

Training

Training is the process of teaching a model using large amounts of data.

Simple Formula

More Data

+

Better Training

=

Better Model

Dataset

A collection of information used to train AI. Examples

- Books
 - Websites
 - Articles
 - Images
 - Code
-

Parameters

Parameters are values the model learns during training.

More parameters generally allow a model to learn more complex patterns, but **quality of data and training** matter just as much.

Token

A token is the smallest unit of text processed by an AI model.

Example

ChatGPT is amazing.

Possible tokens:

Chat GPT

is

amazing

.

Context Window

The amount of information the model can remember during a conversation.

Large Context Window = Better memory.

Temperature

Controls creativity.

Low Temperature

- More accurate
- More consistent

High Temperature

- More creative
 - More varied
-

Hallucination

Sometimes AI gives answers that sound correct but are actually wrong.

Always verify important facts.

Fine-tuning

Training an existing model with additional, specialized data so it performs better on specific tasks.

Example:

A general AI model fine-tuned for legal documents or medical reports.

Quick Revision Box

Model → Learns from data

Dataset → Training information

Parameters → What the model learns

Token → Smallest text unit

Context Window → AI's temporary memory

Temperature → Creativity level

Hallucination → Confident but incorrect answer

Fine-tuning → Specialized training

Transformer Architecture (Most Important)

Interview Definition

A **Transformer** is the architecture behind modern AI models like **ChatGPT, Gemini, Claude, and Llama**. It understands relationships between words using **Self-Attention**, making it faster and more accurate than older models.

Why was Transformer introduced?

Older models (RNN & LSTM) had problems:

Slow training

Forgot earlier words in long sentences

Couldn't process words in parallel

Transformers solved all these problems.

How Transformer Works

Input Sentence

|



Tokenization

|



Word Embeddings

|



Positional Encoding

|



Self-Attention

|



Feed Forward Network

|



Output

Components

Tokenization

Breaks text into small pieces called **Tokens**.

Example

"AI is amazing."



AI | is | amazing | .

Embedding

Converts words into numbers (vectors) so AI can understand them.

Think of it as:

Word → Mathematical Representation

Positional Encoding

AI doesn't naturally know word order.

Positional Encoding tells AI:

- Which word comes first
- Which comes later Without it:

Dog bites man

=

Man bites dog

AI would treat both similarly.

Self-Attention

The most important part of a Transformer.

It helps AI decide **which words are important** while reading a sentence. Example

The cat sat on the mat because it was soft.

AI understands

"It" → refers to **mat**, not cat.

Why Self-Attention is Powerful

Understands long sentences

Finds relationships between words

Processes everything together

Improves accuracy

Why Transformers Beat RNNs

Transformer	RNN
Parallel processing	One word at a time
Faster	Slower
Better memory	Memory fades

Transformer	RNN
Handles long text	Struggles with long text

Remember

Transformer = Brain of every modern LLM

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Large Language Models (LLMs)

What is an LLM?

LLM = Large Language Model

A neural network trained on billions of words to predict the next token.

Simply:

Read → Learn Patterns → Predict Next Word

Examples

Model Company

GPT OpenAI

Gemini Google

Claude Anthropic

Llama Meta

Mistral Mistral AI

How an LLM Works

Massive Dataset



Training



Transformer



Predict Next Token



Response

Training Data

LLMs learn from:

Books

Websites

Articles

Source Code

Research Papers

Inference

Training = Learning

Inference = Answering questions

Most users only interact during **Inference**.

Context Window

The amount of information AI remembers in one conversation.

Large Context Window means:

Better memory

Longer conversations

Better summaries

Temperature

Controls creativity.

Low (0–0.3)

- More factual
- Less creative

Medium (0.5–0.8)

- Balanced

High (0.9–1.5)

- Creative
 - Less predictable
-

Hallucination

Sometimes AI generates information that sounds convincing but is incorrect.

✓ Always verify facts for:

- Medical advice
 - Legal information
 - Financial decisions
-

Interview Tip

LLMs don't "know" facts—they predict the most likely next token based on patterns learned during training.

Modern AI Concepts

Embeddings

Embeddings convert words, sentences, or images into **vectors (numbers)** that capture meaning. Example

King

Queen

Prince

Princess

These concepts are stored close together because they have similar meanings.

Vector Database

Stores embeddings instead of plain text.

Purpose:

Faster search

Better semantic search Examples:

- Pinecone
- Weaviate
- ChromaDB
- FAISS

RAG (Retrieval-Augmented Generation)

Instead of relying only on training data, AI first searches trusted documents and then answers using those results.

Question

|



Search Knowledge Base

|



Relevant Information

|



LLM Generates Answer **Why**

Use RAG?

More accurate

Uses latest information

Reduces hallucinations

Fine-Tuning

Improves a pre-trained model with specialized data.

Examples:

- Medical AI • Legal AI
- Finance AI
- Customer Support AI

AI Agents

An AI Agent can:

✓ Think

✓ Plan

✓ Use tools

✓ Perform multiple steps automatically Example:

"Book my meeting, summarize emails, and create a presentation." The AI plans and completes each task.

Open vs Closed Models

Open Models Closed Models

Llama	GPT
Mistral	Gemini
DeepSeek	Claude

Open Models

- Can often be downloaded and customized.
- Good for developers and research.

Closed Models

- Managed by companies through cloud services.
- Usually easier to use with powerful infrastructure.

Quick Revision

Transformer = Architecture behind modern AI

Self-Attention = Understands word relationships

Embedding = Text → Numbers

Context Window = AI Memory

LLM = Predicts next token

RAG = Search + LLM

Fine-Tuning = Specialized training

AI Agent = AI that can plan and perform tasks

Interview Questions (Practice) Q1.

What is a Transformer?

Answer: A neural network architecture that uses self-attention to process text efficiently and understand relationships between words.

Q2. What is Self-Attention?

Answer: It allows the model to focus on the most relevant words in a sentence when generating an output.

Q3. What is an LLM?

Answer: A Large Language Model trained on massive text datasets to predict the next token and generate human-like responses.

Q4. What is the difference between Fine-Tuning and RAG?

- **Fine-Tuning:** Retrains the model with specialized data.
- **RAG:** Retrieves external information at query time without changing the model.

Q5. Why are Transformers better than RNNs?

- Parallel processing
- Better handling of long context
- Faster training
- Improved performance on language tasks

AI Ethics, Responsible AI & Privacy

Definition

AI Ethics ensures AI is **fair, safe, transparent, and beneficial** for everyone.

Why Ethics Matter?

Without ethics, AI can:

- Give unfair decisions
 - Leak personal information
 - Spread fake content
 - Be misused
-

Bias

Bias means AI produces unfair or unbalanced results because of biased training data.

Example:

- Hiring AI preferring one gender
- Face recognition working better for one group than another

Remember

Bad Data → Biased AI

Privacy

Never upload:

Passwords

OTPs

Bank Details

Aadhaar/PAN

Confidential Company Data

Transparency

Good AI should explain:

- Why it gave an answer
 - Where information came from
 - Confidence level (when possible)
-

Copyright

AI-generated content may still involve copyright considerations.

✓ Always:

- Credit original creators when required
 - Check commercial usage rights
 - Verify images and code licenses
-

Deepfakes

AI can create fake:

- Videos
- Images
- Voices

Always verify before believing or sharing.

Responsible AI Checklist

✓ Fair

- ✓ Safe
 - ✓ Secure
 - ✓ Transparent
 - ✓ Human Supervision
 - ✓ Privacy First
-

Interview Tip

AI should **assist humans**, not replace human judgment in critical decisions.

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AI Safety & Security

Best Practices

Verify AI answers

Protect personal data

Use trusted AI platforms

Enable Two-Factor Authentication (2FA)

Keep software updated

Common AI Risks

⚠ Hallucinations

⚠ Deepfakes

⚠ Phishing

⚠ Fake websites

⚠ Data leaks

AI Do's

- ✓ Write emails

- ✓ Learn concepts
 - ✓ Generate ideas
 - ✓ Summarize documents
 - ✓ Coding assistance
 - ✓ Brainstorming
-

AI Don'ts

- Blindly trust every answer
 - Upload confidential files
 - Depend on AI for final legal or medical decisions
 - Share sensitive business data
-

Verify Before Using

For important work, always check:

- Official Documentation
- Research Papers

 Government Websites

- Domain Experts
-

Strong Password Tips

- ✓ Long passwords
 - ✓ Password manager
 - ✓ Different password for every account
 - ✓ Enable 2FA
-

Golden Rule

AI is your assistant, not your replacement.

Think first.

Use AI second.

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AI Applications & Future

AI is Used In

Software Development

Healthcare

Banking

Education

E-commerce

Marketing

Transportation

Manufacturing

Gaming

Media

Future of AI

Upcoming trends:

AI Agents

Personal AI Assistants

AI Search

AI-powered Coding

Robotics + AI

Multimodal AI (Text + Image + Audio + Video)

AI Agents

AI Agents can:

✓ Plan

✓ Reason

✓ Use tools

✓ Complete multi-step tasks Example:

Research → Create slides → Write email → Schedule meeting

Skills to Learn

✓ Prompt Engineering

✓ Python

✓ APIs

✓ SQL

✓ AI Tools

✓ Cloud Basics

✓ Git & GitHub

✓ Communication

Human Skills Still Matter AI

cannot fully replace:

- Creativity
- Leadership
- Ethics
- Emotional Intelligence
- Critical Thinking
- Decision Making

Career Opportunities

- AI Engineer
- Prompt Engineer
- Data Scientist

- ML Engineer
- AI Product Manager
- AI Consultant
- AI Researcher

Remember

People who know **how to use AI effectively** will have a significant advantage.

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Final Cheat Sheet (5-Minute Revision)

Core Concepts

AI → Machines performing intelligent tasks

ML → AI learns from data

DL → Uses neural networks

Generative AI → Creates new content

LLM → Predicts the next token

Prompt Engineering

Context

Role

Action

Format

Tweaks

→ CRAFT Framework

Model Terms

Dataset → Training data

Training → Learning process

Parameters → Learned values

Token → Smallest text unit

Embedding → Numerical meaning

Context Window → Temporary memory

Temperature → Creativity level

Transformer

Architecture behind modern LLMs

Uses Self-Attention

Faster than RNNs

Better long-context understanding

Self-Attention

Helps AI understand:

Which words are important?

LLM

Examples:

- GPT
- Gemini
- Claude
- Llama
- Mistral

Modern AI

RAG → Retrieve + Generate

Fine-Tuning → Specialized training

AI Agent → Plans and performs tasks

Ethics

- ✓ Fairness
 - ✓ Privacy
 - ✓ Transparency
 - ✓ Human Oversight
 - ✓ Responsible Use
-

Security

- ✓ Verify AI output
 - ✓ Don't share sensitive data
 - ✓ Use 2FA
 - ✓ Check official sources
-

15 Most Asked Interview Questions

1. What is AI?
2. AI vs ML vs Deep Learning?
3. What is Generative AI?
4. What is Prompt Engineering?
5. Explain the CRAFT framework.
6. What is a Token?
7. What is an Embedding?
8. What is a Transformer?
9. Explain Self-Attention.
10. What is an LLM?
11. What is a Context Window?
12. What is Hallucination?
13. Difference between RAG and Fine-Tuning?
14. What are AI Agents?

15. What are the ethical challenges of AI?

Memory Trick

AI

|

|— ML

|

└─ DL

|

└─ Generative AI

|

└─ LLM

|

└─ Transformer

|

└─ Self-Attention

|

└─ Tokens

|

└─ Embeddings

|

└─ Context Window

One-Line Summary

AI learns from data, Transformers help it understand language, LLMs generate responses, Prompt Engineering improves results, and humans remain responsible for using AI ethically and safely.